

## VESLYANA, Russian Federation

WMO: 237010

Lat: 62.9985N

Lon: 50.8983E

Elev: 338

StdP: 14.52

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -34.5      | -28.8 | -39.9                         | 0.6 | -34.0 | -34.4 | 0.8 | -28.8 | 14.5                  | 15.1 | 12.6 | 16.1 | 1.5                      | 0    | 0.653 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.4                   | 82.5            | 65.7 | 78.6 | 63.6 | 75.0 | 61.9 | 68.1                | 77.9 | 66.0 | 75.1 | 64.0 | 71.9 | 5.1                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.8                            | 93.1 | 72.2 | 62.7 | 86.1 | 69.4 | 60.5 | 79.8 | 68.6 | 32.6          | 78.0 | 30.9 | 75.0 | 29.4 | 72.1 | 74.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 14.6          | 12.3 | 11.4 | DB                         | -40.2 | 86.6               | 6.1 | 2.7                                                | -44.6 | 88.5       | -48.2 | 90.1       | -51.6 | 91.6       | -56.1 | 93.6 | (4) |
| (5)               |      |      | WB                         | -39.7 | 70.5               | 7.3 | 2.4                                                | -45.0 | 72.3       | -49.3 | 73.7       | -53.4 | 75.0       | -58.7 | 76.7 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual                           | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|----------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)                              | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                            | 32.2  | 2.4   | 6.7   | 18.6  | 31.7  | 44.9  | 55.3  | 60.7  | 54.7  | 45.9  | 33.7  | 20.0  | 10.6  | (6)  |
| (7)  |                                                                                  | DBStd                            | 22.56 | 16.98 | 15.94 | 11.70 | 8.77  | 9.28  | 8.15  | 7.04  | 7.49  | 6.45  | 7.27  | 13.10 | 17.09 | (7)  |
| (8)  |                                                                                  | HDD50                            | 7289  | 1476  | 1213  | 974   | 551   | 218   | 36    | 5     | 37    | 153   | 506   | 899   | 1221  | (8)  |
| (9)  |                                                                                  | HDD65                            | 12025 | 1941  | 1633  | 1439  | 1000  | 625   | 307   | 169   | 331   | 574   | 971   | 1349  | 1686  | (9)  |
| (10) |                                                                                  | CDD50                            | 808   | 0     | 0     | 0     | 1     | 60    | 196   | 338   | 182   | 30    | 1     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD65                            | 66    | 0     | 0     | 0     | 0     | 2     | 17    | 37    | 10    | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH74                            | 909   | 0     | 0     | 0     | 0     | 69    | 275   | 466   | 99    | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH80                            | 202   | 0     | 0     | 0     | 0     | 12    | 75    | 96    | 19    | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg                            | 5.3   | 4.6   | 5.0   | 5.7   | 5.8   | 5.9   | 5.7   | 5.0   | 5.0   | 5.2   | 5.6   | 5.1   | 5.0   | (14) |
| (15) | Precipitation                                                                    | PrecAvg                          | 23.3  | 1.5   | 1.1   | 1.3   | 1.3   | 1.9   | 2.6   | 2.3   | 3.1   | 2.2   | 2.3   | 1.9   | 1.6   | (15) |
| (16) |                                                                                  | PrecMax                          | 28.9  | 2.3   | 2.3   | 2.5   | 3.0   | 3.5   | 5.2   | 4.0   | 6.2   | 4.0   | 4.8   | 3.2   | 2.7   | (16) |
| (17) |                                                                                  | PrecMin                          | 19.3  | 0.9   | 0.4   | 0.2   | 0.1   | 0.7   | 0.8   | 0.7   | 1.4   | 0.8   | 0.4   | 0.4   | 0.7   | (17) |
| (18) |                                                                                  | PrecStd                          | 2.5   | 0.4   | 0.6   | 0.6   | 0.7   | 0.7   | 1.2   | 0.9   | 1.1   | 0.8   | 1.1   | 0.6   | 0.6   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                             | DB    | 32.7  | 35.4  | 46.6  | 62.1  | 82.1  | 87.6  | 86.8  | 83.5  | 69.6  | 53.0  | 43.7  | 37.2  | (19) |
| (20) |                                                                                  |                                  | MCWB  | 31.7  | 33.8  | 37.4  | 48.4  | 60.1  | 67.5  | 67.2  | 68.1  | 59.8  | 51.2  | 42.5  | 36.4  | (20) |
| (21) |                                                                                  | 2%                               | DB    | 30.6  | 32.9  | 39.9  | 53.1  | 74.6  | 81.6  | 82.7  | 76.5  | 64.3  | 49.3  | 37.5  | 33.5  | (21) |
| (22) |                                                                                  |                                  | MCWB  | 30.0  | 31.7  | 33.8  | 42.8  | 57.8  | 65.6  | 65.8  | 64.6  | 56.0  | 47.2  | 36.8  | 32.5  | (22) |
| (23) |                                                                                  | 5%                               | DB    | 27.1  | 30.5  | 36.2  | 48.5  | 69.3  | 76.7  | 79.6  | 72.5  | 60.3  | 46.1  | 35.1  | 32.2  | (23) |
| (24) |                                                                                  |                                  | MCWB  | 26.2  | 29.6  | 32.5  | 39.8  | 54.4  | 61.3  | 64.4  | 62.2  | 54.2  | 44.3  | 34.4  | 31.4  | (24) |
| (25) |                                                                                  | 10%                              | DB    | 23.7  | 27.3  | 33.9  | 44.6  | 64.2  | 72.1  | 76.3  | 68.4  | 56.9  | 43.6  | 33.3  | 30.4  | (25) |
| (26) |                                                                                  |                                  | MCWB  | 22.9  | 26.3  | 31.7  | 37.9  | 52.2  | 59.8  | 63.3  | 60.4  | 52.8  | 42.3  | 32.5  | 29.6  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%                             | WB    | 32.2  | 34.0  | 38.5  | 50.1  | 63.6  | 70.0  | 72.2  | 69.9  | 60.3  | 51.7  | 43.0  | 36.0  | (27) |
| (28) |                                                                                  |                                  | MCDB  | 32.7  | 34.9  | 45.0  | 58.4  | 74.1  | 82.8  | 80.4  | 78.9  | 67.4  | 52.7  | 44.0  | 36.8  | (28) |
| (29) |                                                                                  | 2%                               | WB    | 29.9  | 32.0  | 35.0  | 44.0  | 59.6  | 67.1  | 68.8  | 66.2  | 57.4  | 47.5  | 36.8  | 32.9  | (29) |
| (30) |                                                                                  |                                  | MCDB  | 30.4  | 33.0  | 38.3  | 51.1  | 72.2  | 78.0  | 78.1  | 74.7  | 62.2  | 48.8  | 37.3  | 33.4  | (30) |
| (31) |                                                                                  | 5%                               | WB    | 26.4  | 29.6  | 33.2  | 41.0  | 56.0  | 64.1  | 66.8  | 63.9  | 55.1  | 44.6  | 34.6  | 31.6  | (31) |
| (32) |                                                                                  |                                  | MCDB  | 27.1  | 30.5  | 35.3  | 46.7  | 67.6  | 73.7  | 75.8  | 69.9  | 59.6  | 45.8  | 35.0  | 32.2  | (32) |
| (33) |                                                                                  | 10%                              | WB    | 23.0  | 26.5  | 31.9  | 38.8  | 52.6  | 61.4  | 64.9  | 61.5  | 52.8  | 42.3  | 32.8  | 29.6  | (33) |
| (34) |                                                                                  |                                  | MCDB  | 23.7  | 27.2  | 33.9  | 43.1  | 62.4  | 69.9  | 73.2  | 67.2  | 56.4  | 43.6  | 33.3  | 30.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB                            | MDBR  | 13.4  | 15.2  | 18.3  | 16.6  | 20.1  | 19.6  | 20.4  | 17.2  | 13.8  | 7.7   | 8.6   | 11.5  | (35) |
| (36) |                                                                                  |                                  | MCDBR | 11.1  | 10.5  | 16.5  | 22.5  | 31.7  | 29.0  | 27.4  | 23.8  | 22.7  | 10.7  | 6.5   | 7.4   | (36) |
| (37) |                                                                                  |                                  | MCWBR | 10.7  | 9.7   | 12.4  | 14.2  | 16.7  | 13.8  | 11.9  | 12.2  | 14.4  | 9.1   | 6.3   | 7.4   | (37) |
| (38) |                                                                                  | 5% WB                            | MCDBR | 10.6  | 9.5   | 12.2  | 18.0  | 28.2  | 24.2  | 22.9  | 21.1  | 19.7  | 10.1  | 6.2   | 7.1   | (38) |
| (39) |                                                                                  |                                  | MCWBR | 10.7  | 9.2   | 9.6   | 12.2  | 16.0  | 13.6  | 11.9  | 12.1  | 13.1  | 9.1   | 6.2   | 7.1   | (39) |
| (40) |                                                                                  | Clear-Sky<br>Solar<br>Irradiance | taub  | 0.242 | 0.290 | 0.322 | 0.353 | 0.353 | 0.358 | 0.405 | 0.373 | 0.336 | 0.309 | 0.246 | 0.228 | (40) |
| (41) | taud                                                                             |                                  | 1.951 | 2.046 | 2.067 | 2.138 | 2.417 | 2.458 | 2.325 | 2.393 | 2.482 | 2.342 | 2.038 | 1.929 | (41)  |      |
| (42) | Ebn at Noon                                                                      |                                  | 113   | 196   | 239   | 259   | 274   | 274   | 255   | 254   | 243   | 197   | 112   | 55    | (42)  |      |
| (43) | Edh at Noon                                                                      |                                  | 11    | 23    | 33    | 38    | 32    | 31    | 35    | 30    | 23    | 17    | 10    | 6     | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg                           | 69    | 285   | 716   | 1229  | 1573  | 1649  | 1672  | 1140  | 616   | 242   | 69    | 29    | (44)  |      |
| (45) |                                                                                  | RadStd                           | 8     | 54    | 104   | 139   | 129   | 153   | 142   | 109   | 82    | 30    | 16    | 4     | (45)  |      |

## Historical Trends

|      |                        | Heating    |            | Cooling    |            |            | Degree-Days |            |            |            |            |
|------|------------------------|------------|------------|------------|------------|------------|-------------|------------|------------|------------|------------|
|      |                        | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD50       | HDD65      | CDD50      | CDD65      |            |
| (46) | Station Only           | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A  | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A |
| (47) | Regional (4 neighbors) | +1.40      | N/A        | N/A        | N/A        | N/A        | N/A         | N/A        | N/A        | N/A        | N/A        |

Nomenclature: See separate page